

RADBOUD UNIVERSITY NIJMEGEN



FACULTY OF SCIENCE

Comparing Text Summarization Models for General Medical Text and Radiology Reports

THE STUDY OF DISCHARGE ME! DATASET

THESIS MSc DATA SCIENCE

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Chapter 1

Introduction

1.1 Background Information

To guarantee an effective treatment, medical records are important in healthcare - being the lifeblood of patients. Records contain patient's medical history, diagnosis, treatment, prescriptions, progress and discharge instructions. Having access to the records, doctors can develop precise diagnoses and effective treatment plans. Legal firms can establish relevant facts and backgrounds for a solid case if patients decide to court for ineffective treatments. Nonetheless, it requires time and effort to manage comprehensive and intricate medical data using specialized terminology. An extensive effort is required to navigate those stacks of paperwork, and losing or misplacing important information can have detrimental effects. This is where medical record summaries come into play [1].

Text summarization utilizes Natural Language Processing tasks, entailing the identification and emphasizing the important information. This process is also referred to as document summarization in certain contexts, which can be divided into three major categories: extractive - most relevant information is retained in the summary, abstractive - the results are more perceptive and imaginative, without fully copying the original document, and mixed - the summary can be both extractive and abstractive, while keeping some portions of the original text intact.

The world is living under the advancement of the internet that connects people around the world and allows a global scale of information exchange. This also involves the field of medical sciences, where individuals can easily look up their medical problem and treatments, and contact healthcare providers. On the other hand, medical professionals can access the latest research, and treatment in their specialties. However, the amount of information can quickly build up and can be troublesome for both parties. Hence, summarization techniques are employed to filter the unnecessary information, while preserving the quality of the reports. In the medical domain, summarization provides a succinct overview of a patient's whole medical background. It makes all the important details clear in an organized way to facilitate accurate tracking of medical records.

1.1.1 Problem Statement

Presence by the vast amount of medical documents, the task of summarizing medical records is still underdeveloped. It is becoming more difficult to generate timely and meaningful summaries. Moreover, these models are not capable of summarizing from multiple sources at this time. For instance, there can be a lot of articles online regarding a specific news story, thus it would be helpful if the summarizer could include both fresh and common information. [10]

In the medical domain, the summarization problem is not generalized to regular text only but also poses interesting challenges due to complexities of medical documents such as large vocabularies and contexts and sensitivities of medical domains [12]. In radiology reports, while helping create efficient diagnostic and therapeutic plans that are customized to each patient's unique require-

ments, summarization is even more difficult because of several features and challenges it poses compared to general medical text, some reports in particular written in reading level are too comprehensively advanced for patients. Specifically, they struggle with the report lengths, along with their use of complex terminologies and languages, leading to "confusion and anxiety" [25].

1.1.2 Relevance of the Study

In recent years, the pre-training language model has been successful in NLP tasks such as inference and paraphrasing, predicting relationships between texts at a holistic or token level. The BERT model - fine-tuned with only one additional output layer from the base Transformers model - which is superior in machine translation tasks with its state-of-the-art attention mechanism, generalizing well to other tasks [29], is powerful in texts relationships prediction in metrics such as GLUE score or MultiNLI. [7]. From these two baseline architectures, advances in large pre-trained language models, such as BART, GPT-3, and T5, have been significant over several downstream tasks in natural language processing. Those model primarily depends on self-supervised pretraining on large corpora of general text. While they are exceptional with general text, their performance in the domain of specialized text, such as clinical or radiology, is rather low. There is a lack of resources to develop text summarization (TS) models, such as "professionally simplified news article" or "parallel document-level TS datasets" [26]. Summarization of important parts for radiology reports leads to time-consuming and erroneous results, as proved by high statistical metrics but factually incorrect summaries [5]. Therefore, NLP-related downstream applications are explored by domain-specific language models. Recently, generative language models have been adapted for tasks such as document summarization. However, in the clinical domain, there are still unique challenges due to their special linguistic characteristics, making clinical text different from the standard language. A study of the T5 model trained on domain specialized text - ClinicalT5 has shown that the model greatly outperforms regular T5 and compares closely to the Unified Medical Language System baseline [19]. Besides T5 and Clinical T5, BART - Bidirectional and Auto-Regressive Transformers, gave promising results being trained on the Cochrane dataset consisting of titles and abstracts of the related clinical trials. This is shown in LexRank in terms of ROUGE and factuality metrics [22].

Given those models, it is important to make comparisons to identify the ones that stand out given the context or whether different models are more suited for summarization tasks. Those comparisons will illustrate the differences in performance and capability of the models in medical and radiology texts. These can be done via performance metrics such as BLEU, ROUGE, and BERTScore [22] to evaluate summaries qualities, domain versatility - how well each model handles different content of texts (medical only versus medical and radiology), computational efficiency - how long it takes to train the models and how much resources are required, qualitative analysis - how adaptable are the models with nuances in medical terminology and abbreviations.

1.1.3 Research Objective and Questions

Given the discussed background and relevance of the study, our objective is to compare the models BART, T5, and its variant (ClinicalT5) about their effectiveness in summarizing text in medical domains. This project is an approach to a challenge on Codabench, in which participants are given a clinical dataset, and are tasked with generating the two important "Brief Hospital Course" and "Discharge Instructions" sections in discharge summaries; to facilitate electronic health record (EHR) creation for patients.

We consider the below research questions:

1. How do ClinicalT5 and mBART perform in summarizing general medical text and radiology reports?
2. How does the addition of NLP methods influence the performance of these models?
3. What is the best input representation for modern summarization models?

1.1.4 Thesis Structure

This thesis report will discuss the relevant literature review in Chapter 2. This includes an overview of the models considered and how they were trained and evaluated on summarization tasks. Next, the third chapter describes the NLP approaches regarding the input text. The fourth chapter will further describe the approaches as implemented methods. The experiment results are described in chapter five. Finally, the sixth chapter will discuss the outcomes, impact, and reflection of the study, along with its benefits and limitations.

Chapter 2

Relevant Work

2.1 Text Summarization Models

Dating back to 1958, text summarization using NLP methods was established and since then has been integrated into the medical domain. Since then, many approaches for document summarization have been proposed and widely applied in various ranges of these domains. Datasets of clinical notes that are friendly toward patients are generated by observing various NLP simplification operations [26]. For radiology reports, sequence-to-sequence learning is implemented in summarizing these reports into impression statements and proposing custom models optimized for this background information [32]. There are 2 types of summarization in this domain: Extractive and abstractive. The former emphasizes the sentences in the document that best captures the text content and rearranges them in an understandable way. However, in medical documents, it is an abstractive approach - to generate novel sentences or paraphrase the whole paragraph that seems more effective. [15]. At Radboud University, there has been an ongoing development of SimpleRad, a web application that explores Dutch radiology reports and provides patient-friendly information about the technical jargon and several important findings such as named entity recognition, observation frequency estimation, and summarization. The concept database is built from Nictiz’s Dutch translation of SNOMEDCT - a systematically organized collection of medical terms and other Dutch healthcare sites. An interface to (abstractive) summarization models is offered via the summarizing module. The main goal is to build a summarizing model - based on the ‘mart-large-50’ checkpoint, fine-tuned on Dutch Radiology reports that produces simplified summaries. Alternatively, we have models that, by using the information entered in the radiological report, will automatically produce results. Hence, the thesis will describe relevant, well-known models such as mBART and ClinicalT5 that can contribute to this task, given the Discharge Me! dataset. [6]

2.1.1 Bidirectional and Auto-Regressive Transformers

BART is a denoising auto-encoder based on a sequence-to-sequence model that applies to a variety of end tasks. It is pre-trained with a denoising auto-encoder objective, which has two stages: the text is noised with an arbitrary noising function (words being shuffled or masked), and then a sequence-to-sequence is trained to rebuild the original text. It combines and generalizes some state-of-the-art NLP models such as BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer). The former is designed to understand context on both sides since the document is encoded bidirectionally, while the latter is suitable to predict the next token in a sequence. This setup is advantageous because arbitrary transformation can affect the text, such as modifying its length. The model is effective in various NLP tasks, including text generation, comprehension, and translation. Specifically, in sequence generation tasks, information is copied and manipulated from the input, then the output is generated autoregressively by the decoder. The model architecture is described in Figure 2.1. [17]

In the context of the medical domain, the BART model is evaluated on two datasets: Cochrane

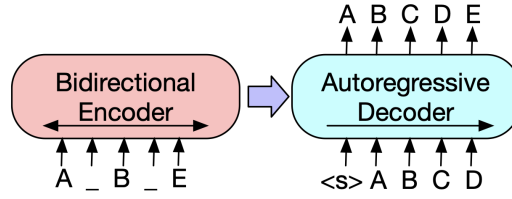


Figure 2.1: BART model: The noisy text is created by substituting parts of an original text with a mask symbol. Then, it is processed by the encoder, analyzing the text from both directions. The decoder tries to predict word-by-word from the encoded text.

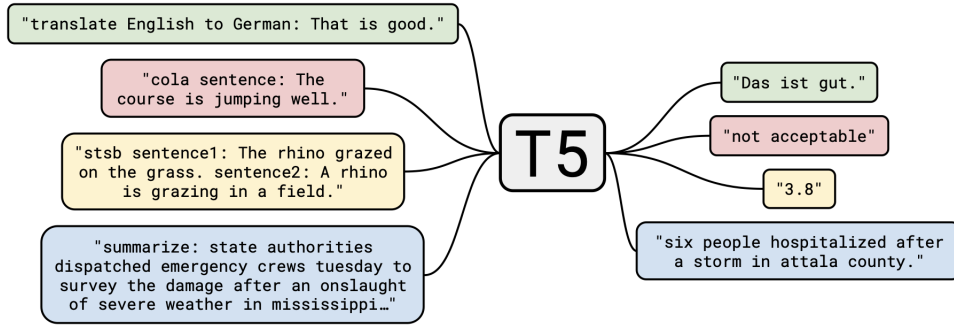


Figure 2.2: The T5 framework, simplifying various NLP tasks as text-to-text format. This allows the same setup - modeling, loss function designing, and hyperparameters tuning, to be used across different tasks

from the Cochrane Collaboration, which contains systematic reviews of all trials from a given clinical question. In this dataset, the inputs to the model are the set of titles and abstracts of the clinical trials, while the target is the conclusions of the reviews. The second dataset is MS^2 , each sentence in a review is classified as background, target, or other and aggregated together. The task is about utilizing the base BART in multi-document summarization, which begins with using unsupervised algorithms: TextRank and LexRank for text extraction. Since the base model is incapable of inputs larger than 1024 tokens, a two-step summarization process was utilized. The first step is to select the most relevant sentences from the original document using TextRank and LexRank principles. Using the BART model, the second step is to generate a cohesive summary from the selected sentences. The results show that the Cochrane dataset is better than the ones on MS^2 , with all the extractive methods being the contributing factors. Overall, the summaries need improvement in the area of complexities of medical texts. [22]

2.1.2 T5 and its variants

- Baseline T5

Similar to BART, the Text-to-Text Transfer Transformer (T5) model - whose methods involve reframing all NLP problems in text-to-text format, is used for various NLP tasks. Unlike other NLP models that output class labels or text spans, T5 treats every output as text strings. This framework is flexible since researchers can apply the same model, objective, training procedure, and decoding process in every task [24]. The framework along with the model architecture is described in Figures 2.2 and 2.3. The model is trained on a robust foundation - the dataset "Colossal Clean Crawled Corpus" (C4), an extensive pre-trained and clean dataset from Common Crawl. In the summarization aspect, the paper gives in-depth insights into the model's configuration - utilizing both the encoder - to understand the entire context and the decoder - generate a compact summary, training on High-quality data - helping the model with the nuances of producing succinct summaries while maintain-

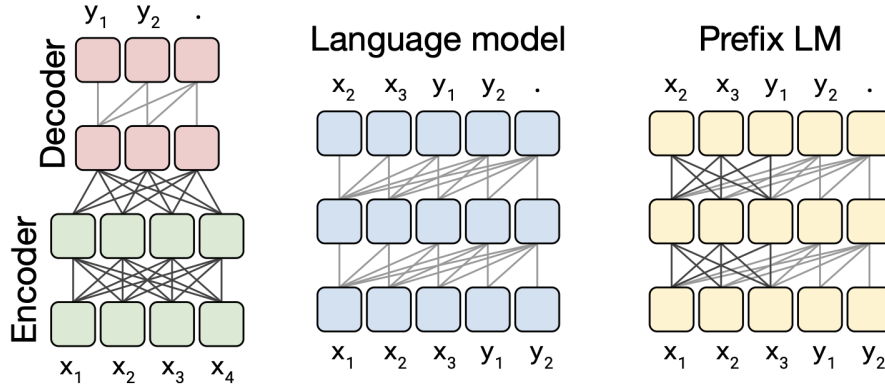


Figure 2.3: Different variations of Transformers: Left - Encoder decoder that uses fully visible masking in the encoder and the encoder-decoder attention, with causal masking in the encoder, Middle - single Transformer stack with input-target concatenation and causal masking, Right - Prefix addition that allows fully visible masking over the input. Dark grey lines mean full visible masking, whereas light grey lines mean causal marking

ing accuracy and coherence, fine-tuning for Medical domain needs - adjusting the model’s parameters to capture the necessary terminology nuances, evaluation of Summary quality - evaluating the model on standard metrics such as ROUGE (Recall-Oriented Understudy for Gisting Evaluation), and measuring the similarities of n-grams between the reference and the generated summaries. The metrics quantitatively assess the summarization ability of the model with different types of content.

- ClinicalT5

In the medical domain, especially clinical, due to the lack of domain-specific generative models, the NLP approaches such as casting non-generation tasks as generative problems are still under development. To bridge the gap, the baseline T5 model is adapted to the clinical setting by training with clinical text. It is trained on the MIMIC (Medical Information Mart for Intensive Care)-III dataset containing textual notes with minimal processing. Then the model is evaluated across several clinical datasets such as UMNSRS-Sim and UMNSRS-Rel, illustrating superb capabilities over T5 baseline across tasks such as summarizing clinical, diagnostic reports and patient treatment notes. It is evaluated against T5 both intrinsically - on model architectures and pre-training procedures, and extrinsically - on Clinical text for document classification, named entity recognition, and natural language inference (NLI) tasks, these evaluations highlight the model adaptability to many clinical NLP tasks [?].

2.2 Medical and radiology text datasets

There are various medical and radiology datasets available which are similar to the Discharge Me! dataset. The newer version of the MIMIC-III used to train the ClinicalT5 model is adapted by Discharge Me!, including a collection of emergency department (ED) admissions. The dataset only needs minimal processing, involving the removal of irrelevant tokens and characters to be ready for training ClinicalT5. Besides the MIMIC datasets, the i2b2 is a notable candidate that contributes significantly to medical NLP research. Specifically, the 2010 i2b2/VA Challenge and 2012 i2b2 Challenge are used to evaluate the ClinicalBERT model. The first one is about extracting medical concepts and relations from clinical text and contains the patient’s medical status, smoking habits, and problems. The second one contains excerpts from clinical narratives and involves medical events timeline identification [16]. Besides i2b2, the BioNLP/N2C2 Datasets were widely used in the medical domain, for example, in the 2018 and 2019 n2c2 challenges. Those challenges involve standardizing clinical concepts for Medical Concept Normalization, supported by clinical

notes annotated with UMLs; and extracting concepts and classifying text from discharge summaries in the identification of patient smoking status task.

Besides medical, there were also radiology-specific datasets that aided in the development of medical context summarization models. Notably, the PadChest containing over 160000 X-ray radiology images annotated with findings and reports, was used in the paper with the same name and a paper [14], and another one that improves chest X-Ray classification by combining data-driven features with radiology-informed knowledge [3].

2.3 Evaluation Methods

Various evaluation methods are being carried out in experiments with NLP models, they are important to measure whether the generated text is relevant. Those methods mainly fall into two categories: Intrinsic and Extrinsic Evaluations, they range from semantic coherence and lexical similarity.

2.3.1 Intrinsic evaluation

When evaluating the created text’s coherence, grammaticality, and fluency, intrinsic evaluation is important. Human judgment and qualitative evaluations are often incorporated into intrinsic methods, and they are essential for picking up on nuances that automated metrics might ignore. The following quantitative metrics are often used when assessing NLP models:

- **BLEU**: Bilingual Evaluation Understudy - considers how similar the generated and the reference text are by the overlap of n-grams. Before being adapted for summarization tasks, it was used to evaluate machine translation. [23]
- **ROUGE**: Stands for Recall-Oriented Understudy for Gisting Evaluation, this methods evaluate the summary quality by measuring the tokens overlap against one or more reference summaries. ROUGE-1, ROUGE-2 and ROUGE-L are variations that measure the overlap in terms of n-grams and longest common subsequence. The value increases proportionally with the score. [22]
- **BERT-Score**: Considers generated contextual embeddings coming from pre-trained transformers such as BERT instead of computing exact matches. The maximum cosine similarities between the standard tokens (Recall) and generated tokens (Precision) are summed and normalized, coming after getting the similarities among each pair. Similar to ROUGE, higher values mean better scores. [22]
- **Meteor**: Designed specifically to assess the quality of machine translation by comparing synonyms, streaming, and paraphrases between generated and reference text. Compared to BLEU, it is more flexible and focuses more on text meaning.
- **AlignScore**: New evaluation metric that considers semantic similarity and information coverage. It is helpful for summarization tasks that require retaining information from the original text.
- **MEDCON**: Medical concept evaluation metric to measure consistency and accuracy of clinical concepts by calculating F1-score to determine similarities between UMLS concepts found in generated and reference clinical text. [31]

Besides quantitative metrics, qualitative analysis methods such as the following might provide better insights:

- **Human Evaluation**: By using human judges to rate the quality of summaries in their fluency, relevance, coherence, and informativeness, this method can be set up as more free-form commentary or as a scoring system. Human-based assessment is said to be more reliable as it conveys more aspects of a model, thus complimenting automatic evaluation. [11]

- Error Analysis: Manually inspecting the summaries to identify common flaws (such as factual inaccuracies, grammatical mistakes, or missing important details) might yield valuable information for improving the model.

2.3.2 Extrinsic evaluation

Contrary to intrinsic evaluation which assesses solely the model’s output, extrinsic approaches assess the model performance in a larger context - such as the final target of a process or in realizing real-world tasks. The key methods for extrinsic evaluation include User studies, Integration with Large Scale Applications, A/B testing, and horizontal studies. The first approach is about the effectiveness of the summarized texts for users in specific tasks, including task completion time or task accuracy - with the help of the summaries. The second approach means that the output summary performance is analyzed as input for other systems or an operational workflow, such as a Clinical Decision Support System (CDSS). The third concept implies different groups of users being exposed to summarized content and measuring the effect on their efficiency or satisfaction over the original content. Lastly, horizontal studies involve observing the effect of summarizations over time, about changes in the domain they are applied to.

In the context of medical summaries, there have been various works to evaluate the effectiveness and efficiency of the summaries in supporting clinical decisions, improving patient care, and reducing document complexities. First, the studies by Zhao, Y., et al claim that using the summarized text in patient records enhances the speed and accuracy of clinical decision-making, allowing more time to take care of patients. Second, the work by Meystre et al. shows that EHR systems show a reduction in document processing time and improved accuracy after the integration of summarization tools. [20]

Chapter 3

Approach

In order to prepare the data as input to mBart and ClinicalT5 model, we will introduce how the CSV files in the dataset are processed using relational database concepts by each considering the files as tables having primary keys and foreign keys, then using NLP methods for the input texts. After that, we will outline evaluation methods for the model, including but not limited to the intrinsic quantitative metrics.

3.1 Data handling and processing

In this study, we use the "Discharge Me!" Dataset from Stanford containing the discharge summary and the goal is to generate the two critical sections "Brief Hospital Course" and "Discharge Instructions" in discharge summaries. The dataset contains a total of 109,168 samples - emergency department admissions, which is split into a training set (68,785 samples), a validation set (14,719 samples), a phase I testing set (14,702 samples), and a phase II testing set (10,962 samples). The dataset includes multiple linked tables, namely 'discharge', 'discharge_target', 'radiology', 'diagnosis', 'edstays', and 'triges'. Each table holds important unique identification and medical information for constructing comprehensive patient discharge summaries. The tables in the dataset are visualized in an Entity Relational Diagram (ERD) in Figure 3.1.

The data integration is achieved through a chain of joint operations. For example, the 'discharge_target' table is joined with 'discharge' using the 'hadm_id' field - identifying each hospital visit, creating one of the inputs and two target outputs for the models. This merge facilitates the aggregation of comprehensive medical text with corresponding target summaries. The tables 'edstays' and 'radiology' are merged using a similar joins method to append radiological finding(s) corresponding to each hospital stay.

Since the input texts cannot contain content of the target predictions, sensitive and non-essential information within the discharge summaries is simply masked in the input to focus on clinical content. The masking process not only complies with the requirements but also refines the input data for NLP tasks. Then, several techniques are implemented to linguistically optimize the input, such as sentence simplification - removing common stop-words and punctuation and keyword extraction - using word embedding to highlight important medical terms. For the baseline comparison, summarization using information retrieval techniques - selecting sentences based on their similarities to a reference sentence is used to condense lengthy text into concise hospital courses and discharge instructions. The examples of several NLP techniques are presented below.

After these pre-processing steps, the data is aggregated into a single structured text segment using separation tokens for each specified component text. These tokens identify different sections of the summary, such as diagnoses, radiology text, and discharge instructions. This structured approach maintains clarity and coherence in the generated summaries for healthcare providers and patients.

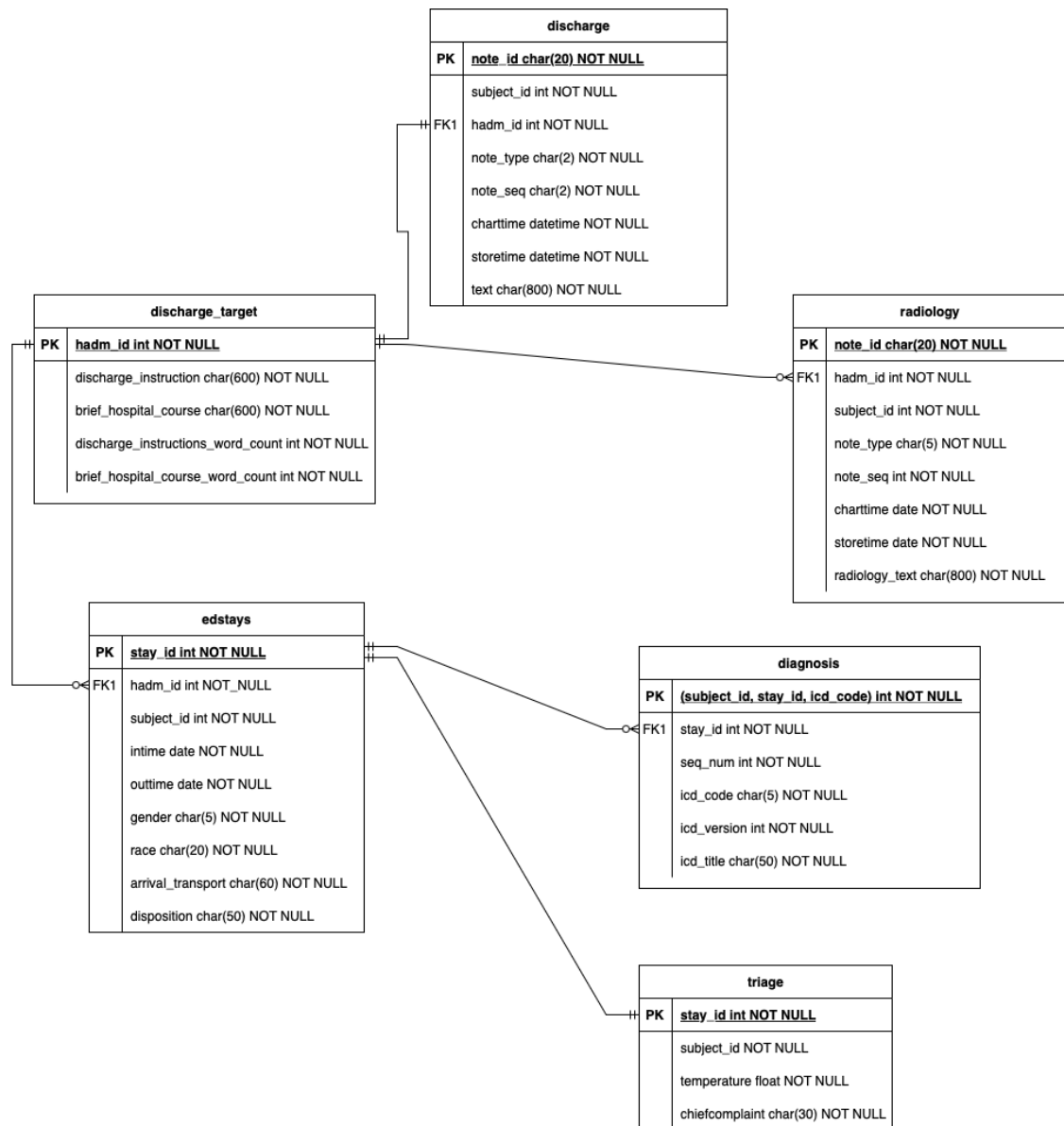


Figure 3.1: The Discharge Me! dataset visualized an Entity Relational Diagram, here the relationships between tables are visualized according to ERD standards

Sentence Simplification

Original: The patient admitted to the hospital with severe chest pain was diagnosed with acute myocardial infarction.

Summarized: Patient severe chest pain acute myocardial infarction

Keyword Extraction

The patient was admitted for severe dehydration and acute kidney injury. Treatment included IV fluids and electrolyte management.

Keywords: dehydration, acute, kidney, injury, IV, fluids, electrolyte, management

Summarization using information retrieval

The patient, a 65-year-old male, presented with chest pain and shortness of breath. After examinations, he was diagnosed with myocardial infarction and underwent successful angioplasty.

Summarized: 65-year-old male with myocardial infarction treated with angioplasty.

Figure 3.2: Examples of Simplification, keyword extraction, and summarization for various medical sentences

3.2 Evaluation method

The main goal of the evaluation is to measure the statistical accuracy and practical implementability of the generated summarized texts. Specifically, they are there to measure the context’s linguistic correctness, information content, and usability within a clinical decision-support of the text.

As discussed in the previous section, here are the intended metrics and methods used to assess the internal performance of mBART and ClinicalT5 with regards to text summarization, are BLEU, ROUGE, METEOR, BERTScore, Meteor, AlignScore, and MEDCON (section 2.3.1). These metrics measure the quality of summary text quantitatively. Manual methods such as human evaluation and error analysis will ensure that the summaries meet the standards of relevancy and factual correctness.

User studies with medical professionals such as doctors and researchers can be an extrinsic evaluation method. Since it is difficult to consult many medical professionals, the thesis is limited to a semi-automated process, it will simulate experts’ evaluation with AI by using models to assess semantic similarities. Additionally, the author can evaluate the generated summaries and rate the summaries’ structure, grammar, coherence, and clinical relevance by himself. Finally, the feedback is collected and the coverage scores are calculated and compared between generated and reference summaries.

Chapter 4

Method

4.1 Parsing and processing the inputs

Extending the methods used in [22] to preprocess the data, the thesis will implement several NLP strategies. Before that, the inputs are read from the given ‘csv.gz’ files using pandas, creating each dataframe that corresponds to a data table from the dataset. According to the dataset description, the ‘text’ column (later renamed to ‘discharge_text’ in the ‘discharge’ table should be the main input for the models, and ‘discharge_instructions’ and ‘brief_hospital_course’ should be the targets that the models trying to predict. Firstly, to ensure the integrity of the study (that the discharge input texts might contain some part of the targets), the input is masked with a space character (‘ ’) whenever we find an overlap between the discharge instructions and hospital courses in the input, and renamed to ‘masked_discharge_text’. The result of masking is a reduction in the length of the inputs ‘discharge_text’, as shown in Figure 4.1.

Inspired by relational databases management system (RDBMS) concepts, where each table contains a primary key that uniquely distinguishes a specific record and (may) contain foreign keys to connect to other tables, to specify a relationship between them, we combine the ‘discharge_target’ with ‘discharge’ to get the input and targets into one single table. Next, the merged table is merged with ‘ed_stays’ on the common key ‘hadm_id’ to access the multiple stays per hospital visit (‘stay_id’) since the two tables are of one-to-many relationship. Similarly, the merged table is joined with radiology through the ‘hadm_id’, resulting in a hospital visit having at least one radiology report. Following that, the result is then joined with the diagnosis table on the shared ‘stay_id’ to find out what are the patients diagnosed with during each stay, in the form of ICD titles. Finally, a similar process is done with the merged result table and the ‘triage’ table



	discharge_text	masked_discharge_text
count	550635.000000	550635.000000
mean	2153.677483	1674.964269
std	929.308674	793.930865
min	242.000000	132.000000
25%	1517.000000	1140.000000
50%	1997.000000	1550.000000
75%	2591.000000	2048.000000
max	7563.000000	7121.000000

Figure 4.1: Statistics for the length of the input, here, ‘masked_discharge_text’ is the original input text after masking

to determine what the `chiefcomplaint` - denoting a patient's self-diagnosis of their situation, associated with each `stay\_id`. The intermediary dataframe is called `df.cleaned`, in which data is later grouped.

After the tables are joined, we create a table that contains the input, targets, and supporting variables. To combine the rows and avoid repetition of content in each row, the ICD titles and `chiefcomplaint` are merged into the set format, for example, { Chronic obstructive pulmonary disease w (acute) exacerbation, pneumonia }. The radiology reports for each visit are also joined, each report is separated with a special token, namely `</ RAD_SEP>`. They are merged with the `masked\_discharge\_text`, with each section separated with the special token `</ s>`. The resulting dataset is called `df.cleaned`, with the discharge text, ICD titles, chief complaint, and radiologies reports of each hospital visit combined. Finally, since all the contents of stays for each hospital visit (`hadm\_id`) - ICD, chiefcomplaint, and radiologies are merged, the final dataframe - `df.aggregated` is created by grouping the `hadm\_id` to eliminate repetition of records.

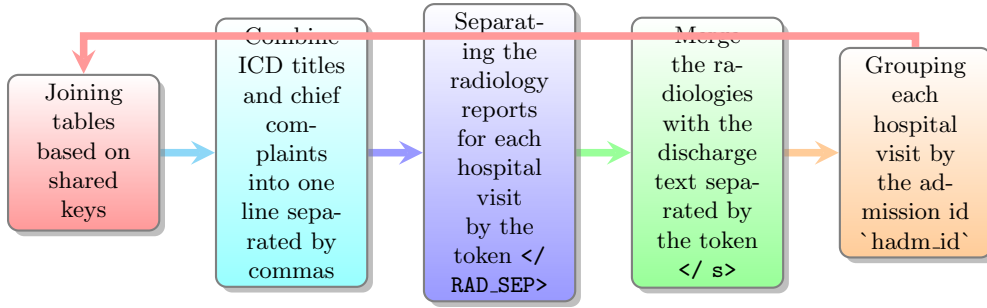


Figure 4.2: The final input dataframe creation process for each record

4.2 NLP methods

Applying different NLP methodologies to medical texts and radiology reports can improve the quality and relevancy of the input before training on NLP specialized models. As discussed in the previous sections, several methods such as sentence simplification, keyword extraction, entity based recognition will be discussed. Although the latter two methods will extend the input by some number of tokens, we believe these will capture the essentials of the input, in which each record is truncated due to the tokens constraint - if we directly tokenize the original text, important information such as diagnosis, treatment, and admission will be lost. The code that implemented these methods will be included in the Appendix.

4.2.1 Sentence Simplification

Sentence simplification is an NLP technique that preserves the meaning of the original sentence while reducing its complexities. In the medical context, this method is useful where sentences can be long and filled with medical jargon. The performance of NLP models can be improved with these concise and readable texts as input. Currently, text simplification in medical domains only focuses on terminologies, either by generating the explanation or replacing them with simpler terms. This oversees complex sentence structure and sophisticated non-medical terms. [28]. Therefore, trivial methods such as removing stop-words and punctuation can make the texts more concise and coherent.

To perform sentence simplification, we adhere to the following steps:

1. Using the nltk package - a pioneering platform for building Python programs to work with complex tasks of language processing, which provides an accessible user interface for more

than fifty corpora and lexical resources [2], we import the ``sent_tokenize`` method to split the sentence into individual tokens and ``stopwords`` corpus to set a list to filter the text on.

2. The custom punctuation is imported from the ``string`` package, and only the period is kept since we are interested in keeping the sentences intact.
3. The actual simplification is first done by tokenizing the sentence and filtering each word if it is not in the stop words and the custom punctuation lists.
4. For the radiology text, first the special characters such as the smaller or equal sign or the brackets are removed since they contain many more medical descriptions that need explanations, and then the rest is simplified as normal.

4.2.2 Keyword Extraction

The improvements in healthcare have led to more detailed records for patients. Nevertheless, most of these records are in unstructured form. The paper by Tsai, Wu, and Yu proposes a system to convert those records from unstructured to structured, in which they implement a keyword phrase extraction to build a vocabulary dictionary, which in turn is used to organize the unstructured examinations in a structured form. [27]. Inspired by this work, this thesis is interested in identifying the main words within a text that contribute the most to the meaning. In the medical setting, this involves extracting the words that are most relevant to the diagnosis, condition, and treatment of a patient. Specifically, it helps highlight the important information in lengthy medical reports.

The experiment performs keyword extraction by combining statistical and information retrieval techniques - the Term Frequency-Inverse Document Frequency (TF-IDF) method. It is defined as the relevancy of a word in a series of corpus in a text, which increases proportionally with the frequency in the text, but its frequency in the corpus decreases. Term Frequency is the number of occurrences of a word t in a single document, while the Inverse Document Frequency is the logarithm of several documents in the corpus (N), divided by the document frequency - the number of documents that contain the term t . TF-IDF is thus a combination of both measures, in which the formula is the product of TF and IDF [9]. This metric is utilized by the ``extract_keywords`` function. The function highlights important information in medical records or other texts by identifying and returning the most significant keywords in the input text using `TfidfVectorizer` from the `sklearn` library. This technique makes sure that the model focuses on the most important portions of the text throughout the preprocessing stages of NLP tasks.

To implement keyword extraction in Python, we propose the following steps:

1. Import the ``TfidfVectorizer`` from the ``sklearn`` library, which is used to convert a collection of documents into a matrix containing the TF-IDF features.
2. Create a vectorizer from the imported function, specifying the number of features (keywords), in this case, we chose 30 keywords for the discharge text section - which translates to around 50-70 tokens, we found that is suitable for the input limit of 512 tokens for the models and with the medical texts being 2000 tokens on average (Figure 4.1). The common English stop words from the text are removed.
3. To do the fit and transform to the input text, we then call the ``fit_transform`` method from the vectorizer on the input text as a list. It returns the TF-IDF matrix from the learned vocabulary and inverse document frequency of the text.
4. The feature array is created from the learned vectorizer by calling `feature_names_out``.
5. The TF-IDF scores are sorted by converting the TF-IDF matrix into an array and sorting in descending order of the TF-IDF score.
6. Return the converted list of the feature names. The list is then prepended to the text as "Keywords: ", followed by the returned list and the separation token `</s>`

An excerpt of the output for a sample text input is given in Figure 4.2.

Keywords: ___, mg, po, patient, 100, pain, blood, 04, 20pm, discharge, prn, 25, acute, clear, bowel, ra, history, home, oxycodone, diarrhea, daily, constipation, process, bilaterally, admission, ng, non, normal, 21, physical </s> Discharge text: \nName: ___ Unit No: ___\n \nAdmission

Figure 4.3: Example of keyword extraction as an input to the model

Entities: HTN, docusate sodium, RA, Diverticulosis, Small bowel containing right inguinal hernia, ovarian mass, UA, pain, VitB12, GU, Allergies, Cataract, Small hiatal hernia, cancer, oxycodone, edema, diarrhea, RX *polyethylene glycol 3350, anemia, hypertension, Lisinopril, fracture, clubbing, LBP, Thiamine, hematoma, iron, diverticulitis, Colace, bowel movements, liver cancer, Q4H, Q8H pain, NAD, rhonchi, Tylenol, organomegaly, osteoporosis, allergic rhinitis, occipital scalp hematoma, incontinence, Gabapentin, Acetaminophen, osteoarthritis, intra-abdominal, acetaminophen, hemolysis, TMJ dysfunction, Glaucoma, tablet(s, presacral edema, umbilical hernia, ataxia, Vertebral plana abnormality, Sodium, irritable bowel syndrome, breast cancer, constipation, cyanosis, fatigue, Aspirin </s> Discharge

Figure 4.4: Example of extracted entities for a given discharge text

4.2.3 Entity Recognition

Similarly inspired by the paper [27], we are interested in constructing a "vocabulary dictionary" that will help the model recognize the important and under-covered segments in the input by identifying and classifying key elements in a text, for example, diseases, medications, or human body parts. This NLP method is important in understanding the structured medical information within the unstructured text. By using a pre-trained NLP library such as spaCy for named entity recognition - giving "real-world" objects a label, medical entities, and their classifications can be extracted and highlighted in the text. This method is effective in facilitating tasks such as text summarization in clinical environments.

The entity recognition process is described as follows:

1. Install the `scispaCy` package containing models in `spaCy` for processing biomedical, scientific or clinical text. Load the `en\_ner\_bc5cdr\_md`, a Named Entity Recognition (NER) model trained on the BC5CDR corpus which can detect an entity in the medical domain either as `DISEASE` or `CHEMICAL` [21].
2. Create a `doc` object by calling the `nlp` pipeline loaded the NER model earlier. This object contains the processed text and the recognized entities.
3. Iterate the recognized entities from `doc.ents`, creating a tuple of the form `(ent.text)`, in which the entity's text is contained.
4. Return the list containing entities and their corresponding label as described, which is later added to the main discharge text as "Entities: ", followed by the entity names, and the separation token </s>

The example of the discussed output is given in Figure 4.3.

4.3 Model and Data Setup

This section details how the data are set up and prepared for training two NLP models: mBART and ClinicalT5. These models are specialized in text summarization and classification of medical reports, specifically discharge text and radiology. Therefore, the configuration of both the data and the model is important to ensure the training is effective.

4.3.1 mBart Model Setup

Multilingual BART (mBART) is a sequence-to-sequence model that applies to text generation tasks. It can also be fine-tuned in summarization tasks. Here, we elaborate on the data setup and training of the mBART model on medical texts.

Model Configuration

1. Tokenizer and model initialization: The tokenizer is imported as `The BartTokenizer` and the model is `BartForConditionalGeneration`. The model is set to the pre-trained ``facebook/bart-large`` for this task, which can effectively capture and generate sophisticated language structures and patterns.
2. Loading the tokenizer: the tokenizer is loaded from the imported pre-trained model, and the tokenizer converts text into readily available tokens that follow the vocabulary and tokenization rules for the mBART model.
3. Loading the pre-trained model: Two instances of the models, `model_course` and `model_instructions` are loaded and created. These two will handle summarization tasks, the former for summarizing the brief hospital course and the latter will take care of discharge instructions.
4. Adding the separation tokens: We first check if the separation token for the text and radiologies section, namely `</ s>` and `</ RAD_SEP>` is available in the tokenizer vocab and add them.
5. Creating the Dataset and Data Loader: The dataset class (``SummarizationDataset``) used for the mBART model consists of the input and target (summarized) text, the tokenizer, and the maximum input and output length. For summarizing the hospital course and discharge instructions, although the maximum number of tokens that mBART can handle is 1024, we set the maximum number of tokens for the target to be 512. This allows consistency and fairness when running experiments on the ClinicalT5 (which can only allow for 512). We are also interested in whether the mBART with the 1024 tokens limit will improve the overall performance, hence a separate experiment be conducted and the results will be documented in the Appendix (table A.1). For each record in the input and the target, they are tokenized and parsed as PyTorch tensors (``pt``). They are then returned as a dictionary containing the ``input_ids``, ``attention_mask``- actual tokens (1) or padding (0). Finally, the `DataLoader` for the training and validation datasets is created from the dataset class. We set the batch size to 4 and shuffle the training data.

4.3.2 ClinicalT5

ClinicalT5 is a variant of the base T5 model, fine-tuned for text generation and summarization in clinical domains. This model is adept in the complex structure and jargon in medical text.

Model Configuration

1. Tokenizer and Model Initialization: For this model, we use the ``T5ForConditionalGeneration`` as the base instance and the ``AutoTokenizer`` for the Tokenizer. The model is set to the pre-trained ``luqh/ClinicalT5-base`` for this task. We set the ``from_flax`` to `True` to be able to load from the pre-trained weights.

2. The rest of the setup is similar to the mart model, except in the ``SummarizationDataset`` class, we add the prefix "summarize: " to the beginning of each input text. This is done by reassigning the input text in the `__getitem__` function at a particular index. Moreover, we set the batch size to 2 since the training task is more memory intensive.

4.4 Model training

This section will elaborate on how the environment for separate models for the hospital course and discharge instructions are set up for training. The setup includes model configuration, optimizers, and learning rate schedulers for training effectiveness. The section will also detail accuracy calculation and training loop. These specified steps guarantee that the models are trained and optimized for medical text summarization tasks.

4.4.1 Optimizer and Scheduler Configuration

For both mBart and ClinicalT5, the optimizers and learning rate schedulers are setup similarly for the course and instructions models. For optimization, we use ``AdamW`` - a stochastic gradient descent method optimizer "based on adaptive estimation of first-order and second-order moments with an added method to decay weights" from the techniques in the paper "Decoupled Weight Decay Regularization" [18]. The parameters and the learning rate of $5e^{-5}$ are passed to the AdamW initialization.

Following the creation of the optimizer, we set the number of epochs to be 3, as the train and validation loss stabilized after 3 epochs, which will be discussed in the appendix. We calculate the total step for the scheduler by multiplying the length of the DataLoader with the number of epochs and initialize the scheduler with the ``get_linear_schedule_with_warmup`` function from the ``transformers`` package, passing the optimizer and the number of steps to the parameters.

4.4.2 Training Loop

The training loop is implemented as normal Deep Learning concepts, that is iterating over the data from the initialized DataLoader, performing forward and backward passes, and updating the model weights per iteration.

Accuracy Calculation function

To calculate the accuracy per batch, we create the ``calculate_accuracy`` function, which measures the accuracy between the model's predictions and the true labels. The function takes the model's output logits and the labels from the same batch and calculates the accuracy as a float by dividing the total number of matches by the number of elements.

Training and Validation Process

The training consists of iterating over batches of data loader, performing backward and forward passes, and using gradient descent to tune model parameters. After each batch, the training loss, train accuracy, and validation loss are logged using the ``wandb`` API to monitor and graph the model's performance.

4.5 Model Output Generation

The section describes how the targets are generated using the trained mBART and ClinicalT5. The steps for data preparation, model output generation, submissions, and reference file generation will be discussed. These steps are important to ensure the outputs match the requirements and the performance of the models is validated.

4.5.1 Data Preparation

Before generating the predictions, the test data, which we are using from the `test\_phase\_2` folder, is prepared similarly to the training data. It includes initializing the tokenizers (`AutoTokenizer`), tokenizing the data and converting to a PyTorch Dataset (`SummarizationDataset`) and creating a DataLoader. For this task, the batch size is set to 16 and the `max\_target\_length` for the dataset is set to 512 for the hospital course and also 512 for the discharge instruction, similar to the training data.

4.5.2 Model Output Generation

For both models, the process involves feeding the test data and generating predictions. The generated output, currently in the format of the token, will be decoded back to normal text readily available for submission output generation. For each batch in the test data loader, the outputs are generated using mixed precision (autocast) to speed up the process. The output is saved using the `pickle` library for further evaluation.

4.5.3 Submission and Reference Files Assemble

Using the loaded `discharge\_target` file from `test\_phase\_2`, we build the reference file by getting the necessary columns (`brief\_hospital\_course` and `discharge\_instructions`), tokenizing using the tokenizer with the separation tokens added to make sure it is of appropriate lengths (512 tokens). Then the outputs are decoded back to normal text. Finally, the dataframe is generated using the pandas DataFrame method, passing the hospital admission ids (`hadm\_id`), the hospital course, and discharge instructions - according to the scoring script requirement; and saved as a CSV.

The submission file is generated similarly, first, the two predicted outputs are "unpickled" from the saved files, and then the similar columns are saved into a dataframe and the submission CSV file.

4.6 Evaluation

The evaluation process involves assessing the quality and relevance of the generated summaries. This section will elaborate on the statistical metrics used for intrinsic evaluation, and the user studies set for extrinsic evaluation.

4.6.1 Baseline creation

We create an extractive summarization as a baseline to compare against the generated summaries by selecting the most important sentences from medical texts and radiology reports. This NLP method ranks sentences based on their relevancy to several selected themes using TF-IDF and Cosine Similarity measures. The latter is a metric that determines the similarities between data objects, irrespective of their size by treating them as a vector. It is calculated by taking the dot product and divided by the regular (length) product of the two vectors [8]. Here, the TF-IDF representations of medical texts are represented as vectors, if the similarity is 1, the two texts are identical and 0 means no similarity. Favoring certain information in a text can guarantee the NLP models are trained on the most important content, thus improving their performance.

The steps involved in this information retrieval technique are as follows:

1. Splitting text into sentences: Here, regular expressions are used to split the text into separate sentences. This step allows each sentence to be evaluated for relevance independently.
2. Vectorizing sentence: As discussed in the "Keyword Extraction" section above, the TF-IDF vectorizer imported from the 'sklearn' library transforms the sentences into numerical vectors, which quantifies the textual data for analysis.

3. Creating reference vectors: The reference sentences, used as "themes" to depict the key concepts, are also vectorized. Here, the references are chosen to be "discharge summary", "patient diagnosis", "treatment plan", "follow-up instructions" and "radiology findings", since we find these to be consistent with the patterns of the input discharge text. These vectors are the standard for finding the relevancy of each sentence in the input text.
4. Calculating Cosine Similarity: The value is calculated between the reference vectors and the input sentences vector, which defines how much the alignment is between each sentence and the references
5. Scoring and ranking sentences: For each sentence, a relevance score from the calculated cosine similarity is assigned. Then, the scores are summed, giving each sentence only one relevance score. The sentences are then ranked in descending order of relevance scores.
6. Selecting the top sentences: The top 20 sentences are selected, we find this number to be optimal assuming each sentence consists of 15-20 words given the summary is between 150 to 450 words out of the original 1500-word text (on average). Generating the output: Finally, the chosen sentences are concatenated, making sure the most important information is retained.

The example of the discussed output with 15 sentences is given in Figure 4.5. It can be seen that the discharge text is significantly shorter, as the separation token `</ s>` can be observed separating each section. Then, the output is split into hospital course and discharge instructions based on several keywords such as 'discharge instructions', 'follow-up', and 'hospital course',... The baseline is then tokenized using the corresponding tokenizer for each model (mBART and ClinicalT5), shortened into 512 tokens, and then decoded to be readily available for evaluation.

4.6.2 Intrinsic Evaluation Metrics

Those metrics quantitatively measure the quality of the generated summaries compared to the reference summaries. The used metrics are:

- BLEU-4: Measuring overlap at 4-gram level, the higher the score, the better the quality.
- ROUGE-1, -2, -L: Measuring overlap on unigrams, bigrams, and longest common subsequence, respectively.
- BERTScore: Measuring similarities based on pre-trained BERT contextual embeddings.
- METEOR: Measuring words' order, synonym, and stemming on the generated text.
- AlignScore: Measuring alignment between the texts using pre-trained models
- MEDCON: Measuring medical relevance on the generated text.

We are given the scoring script from the competition organizer. First, the score for the two target sections (hospital course and discharge instruction) is scored separately. For each metric, the mean across all test samples is calculated. Then, the scores for each of the two targets are averaged and reported as metric names on the scores output. Finally, the overall score is computed by averaging over all metrics. [30]

For example, if there are N samples and s is the score for a given sample, then the mean across all samples of a given metric S is [30]

$$S = \sum_{i=1}^N (s_i) / N \quad (4.1)$$

Then the β denoting the mean of a given metric will be:

$$\beta = (S_{BriefHospitalCourse} + S_{DischargeInstructions}) / 2 \quad (4.2)$$

summarize: \nName: ____ Unit No: ____ \nAdmission Date: ____
 Discharge Date: ____ \nDate of Birth: ____ Sex: F \nService: MEDICINE \nAllergies: \nAspirin \nAttending: ____ \nChief Complaint: \nweakness, diarrhea \nMajor Surgical or Invasive Procedure: \nNone \nHistory of Present Illness: \nMs. ____ is a ____ year-old woman with PMH significant for \nchronic anemia, osteoporosis, hypertension, ataxia, and recent \nL5 fracture in the setting of recurrent falls who presents from \nhome with fatigue and generalized weakness and diarrhea. \n The patient's recent history is notable for the follow: \n - On ____, she presented with 4 days of LBP s/p fall from \nstanding at which time imaging revealed acute L5 fracture. She \nwas evaluated by Spine team who recommended early mobilization, \npain control, but no brace required. She was evaluated by ____, \nand discharged to ____. \n - She was discharged home with ____ on ____. \n - On ____, she again presented to ____ s/p fall from \nstanding while trying to reach for a glass of water. She did \nhave a occipital scalp hematoma, but imaging including ____, \nC-spine CT, and L hip X-ray were negative for acute process so \npatient was discharged home. \n She now represents with generalized fatigue and diarrhea. In \nthe setting of opiates for her L5 fracture, the patient has had \nconstipation (5 days with no BM) for which she took a "natural \nlaxative" the evening prior to presentation. The patient had 2 \nbowel movements in the morning of presentation and one episode \nof incontinence with diarrhea while sleeping. In this setting, \nshe felt very weak and called EMS and was brought to ____ ED. \nIn ____ ED: \n - Initial VS 98.7/72/130/46/16/99% RA \n - Labs notable for Chem-7 with BUN/Cr 36/1.1 otherwise wnl, CBC \nwith H/H 7.8/25.4 (baseline Hct 31), iron studies with iron 49 \nferritin 249, hemolysis labs with LDH 163 Tbili 0.2 hapt 170. \nUA negative. \n - CXR without acute intrathoracic process. CT A/P without acute \nprocess. \n - The patient was administered \n ____ 21:25 PO/NG OXYCODONE (Immediate Release) 2.5 mg \n ____ 21:25 PO/NG Gabapentin 100 mg \n ____ 21:25 PO/NG Thiamine 100 mg \n ____ 21:25 PO/NG Folic Acid 1 mg \n 1u pRBC transfusion started \n - VS prior to transfer 98.5/60/150/56/18/100% RA \nUpon arrival to the floor, VS 149/59/57/18/100% RA. The patient \nis well-appearing and states that she already feels significant \nbetter than when she first presented. \nPast Medical History: \nallergic rhinitis \n anemia \nirritable bowel syndrome \nosteoarthritis \nTMJ dysfunction \nosteoporosis \nHTN \nBilateral Macular degeneration \nGlaucoma \nCataract surgery \nLeft shoulder

(a) Original discharge text

Brief Hospital Course: \n \nTRANSITIONAL ISSUES: \nPatient should have follow up CBC in 1 week to trend Hgb/Hct. Lisinopril 40 mg PO DAILY \nDischarge Disposition: \nHome With Service \nFacility: \nDischarge Diagnosis: \nLaxative-induced diarrhea \nChronic anemia \nDischarge Condition: \nMental Status: Clear and coherent. The patient's recent history is notable for the follow: \n - On ____, she presented with 4 days of LBP s/p fall from \nstanding at which time imaging revealed acute L5 fracture. Lisinopril 40 mg PO DAILY \nDischarge Medications: \n1. Discharge Instructions: \nFollowup Instructions: \nAnemia, unspecified \nWeakness, Diarrhea \nRadiologies-Section: INDICATION: ____ with weakness // r/o infiltrate \nTECHNIQUE: AP and lateral \nCOMPARISON: Chest radiograph dated ____ \nFINDINGS: \nAP and lateral chest radiograph demonstrate stable cardiomeastinal and hilar contours. summarize: \nName: ____ Unit No: ____ \nAdmission Date: ____ Discharge Date: ____ \nDate of Birth: ____ Sex: F \nService: MEDICINE \nAllergies: \nAspirin \nAttending: ____ - Patient needs 24 hour supervision and home physical therapy. The patient \nis well-appearing and states that she already feels significant \nbetter than when she first presented. Patient stated that she had a friend who would provide 24 hour \nsupervision. - Patient was discharged with Colace and Miralax as needed for \nconstipation if still using oxycodone. The patient had 2 \nbowel movements in the morning of presentation and one episode \nof incontinence with diarrhea while sleeping. - Patient should take Tylenol for pain and avoid oxycodone \nunless having severe pain. She did \nhave a occipital scalp hematoma, but imaging including ____, \nC-spine CT, and L hip X-ray were negative for acute process so \npatient was discharged home. In \nthe setting of opiates for her L5 fracture, the patient has had \nconstipation (5 days with no BM) for which she took a "natural \nlaxative" the evening prior to presentation. DISCHARGE PHYSICAL EXAM: \nVital Signs: AF, 150/70, 60, 16, 99% on RA \nO: ____ \nGeneral: Well-appearing elderly woman, appears younger than \nstated age, NAD \nHEENT: Sclera anicteric, MMM, oropharynx clear, EOMI, PERRL, \nneck supple, JVP not elevated, no LAD \nCV: Regular rate and rhythm, normal S1 + S2, no murmurs, rubs, \ngallops \nLungs: Clear to auscultation bilaterally, a

(b) Baseline summary

Figure 4.5: Comparison between excerpt of original and the baseline extractive summary

And finally, the overall system score is calculated by averaging the 8 β values:

$$Overall = \sum_{i=1}^8 (\beta_i) / 8 \quad (4.3)$$

We will get the overall scores of the prediction text from each model trained on each NLP method, and then compare them to the scores produced from the baseline text.

4.6.3 Extrinsic Evaluation

Extrinsic evaluation is about measuring the generated summaries' applicability and relevance in a real-world context. This can be done through the perspective studies of medical professionals. However, since the thesis is an individual project and is mostly interested in the performance of the model rather than improving upon the generated text and applying it to real-world settings, it is difficult to gather reputable professionals for this task. Therefore, this will be done individually for a selected number of summaries as manual validation, with the help of AI tools specialized in semantic similarity.

Simulating Expert Evaluation with AI

The evaluation of generated summaries through content analysis for medical relevance and factual correctness can be done by pre-trained medical models such as 'Bio_ClinicalBERT'. The model can evaluate the semantic similarity between the generated and reference medical text through cosine similarity, the higher the result score, the more relevant and accurate the generated text. This is different from the traditional BERTScore calculation in that the use of 'Bio_ClinicalBERT' will generate embeddings more oriented toward the clinical domain.

The implementation can be done by first loading the tokenizer and the model, in this case, 'emilyalsentzer/Bio_ClinicalBERT'. Next, we tokenize the input and reference texts and get the embeddings through the models. Lastly, we calculate the cosine similarity between the two embeddings for the final score. Due to the computational time limit, we select randomly 1000 records from both the reference and the generated texts.

Self Manual Evaluation

The thesis will evaluate the summaries according to several predefined criteria. This involves assessing the quality of summaries on various language properties such as coherency, completeness, accuracy, and clarity. The evaluation criteria are further detailed below. An example of an evaluation form is in Table 4.1.

- Accuracy: Whether the information in the generated text is factually correct
- Clarity: Whether the information is easily comprehensible
- Completeness: Whether the information contains all necessary contents from the reference
- Relevance: Whether the information is relevant to the reference

This method will utilize a Likert scale (from 1 to 5) with 5 being the highest score to rate each criterion, additionally, we will provide our perspectives on the text. We will randomly select 10 summaries and rate them according to these criteria. Overall, these methods of combining both AI and manual evaluation can well-roundedly assess the quality of the text and performance of the model without the need to have medical professionals.

The code repository that implements these steps is on GitHub, the link is under Appendix A.3.

Table 4.1: Example of Evaluation Table

Criterion	Rating (1-5)	Comments
Accuracy		
Clarity		
Completeness		
Relevance		
Overall		

Chapter 5

Results

5.1 Statistical metrics

This section details the eight discussed statistics above for both the mBart and ClinicalT5 model given their respective preprocessed input methods. These metrics will then be compared to the statistics of the extractive summarization baseline reference.

5.1.1 mBART results

Below are the results for the mBART model on different preprocessing methods

Table 5.1: Results for the mBART model on various preprocessing methods. Here, R stands for the ROUGE metric, BERT is for BERTScore and MET is for METEOR

Preprocessing Method	BLEU	R-1	R-2	R-L	BERT	MET	Align	Med	Overall
Base (no preprocessing)	0.087	0.357	0.115	0.188	0.371	0.274	0.271	0.320	0.248
Sentence Simplification	0.083	0.354	0.113	0.205	0.376	0.273	0.254	0.327	0.248
Keywords Extraction	0.088	0.426	0.208	0.282	0.372	0.275	0.254	0.324	0.279
Entity Recognition	0.090	0.421	0.174	0.244	0.378	0.271	0.245	0.328	0.269

The base method will serve as the baseline performance of the model, and later on will be discussed against the models with preprocessed inputs.

Compared to the base model, the Sentence simplification model has slightly lower BLEU, ROUGE-1, ROUGE-2, and METEOR scores. However, there was a slight improvement in BERTScore and MedCon. The overall score differs by a small margin (0.003), suggesting that simplification may reduce detailed content slightly but make it more readable.

The overall result reveals that keyword extraction gives the best performance across the metrics, as shown in the improved ROUGE score. This shows that extracting keywords using the tf-idf method assisted the model in capturing the most relevant information, which leads to better-generated summaries.

The model with the extracted medical entities also improves model performance, specifically the BERTScore and MedCon, showing that the key medical terms are better captured. However, it just came short of the overall score compared to keyword extraction. Overall, the base model performance is worse than methods that improve the context. To further improve the effectiveness, we can combine both the keyword and entity extraction methods, however, we should limit the number of the output since the tokens limit is 512, as the experiment is interested in.

5.1.2 ClinicalT5 results

The results for the clinicalT5 model on different preprocessing methods are below:

Table 5.2: Results for the clinicalT5 model on various preprocessing methods

Preprocessing Method	BLEU	R-1	R-2	R-L	BERT	MET	Align	Med	Overall
Base (no preprocessing)	0.031	0.240	0.107	0.189	0.224	0.150	0.325	0.241	0.188
Sentence Simplification	0.034	0.275	0.099	0.196	0.151	0.151	0.316	0.219	0.180
Keywords Extraction	0.026	0.111	0.049	0.099	0.214	0.137	0.346	0.218	0.150
Entity Recognition	0.046	0.237	0.097	0.180	0.248	0.179	0.262	0.255	0.188

This model with non-preprocessed input will serve as the baseline performance of ClinicalT5, and later on will be discussed against the models with preprocessed inputs.

Compared to the baseline, the sentence simplification method improves BLEU and ROUGE scores but makes the other scores worse, hence the drop in the overall score. This shows that sentence simplification might cause slight information loss for the input.

Unlike the mBART, the keyword extraction method on CLinicalT5 shows significant drops in all metrics, except for Align, which might not be suitable for this model, where summarization might not be efficient in medical text with keywords.

The model with the extracted medical entities using scispacy ‘en_ner_bc5cdr_md’ model and prepend to the input improves the BLEU, BERTScore, METEOR, and MedCon. However, The Rouge and Align scores are worse. This suggests that the most important information is retained, in exchange for the reduction of the text quality. Overall, we should mostly consider this method for ClinicalT5, while Sentence Simplification loosens the critical information, which might be a result of a lack of coherency. Finally, we should not consider keyword extraction for this model, as shown in the worst performance in all metrics.

5.1.3 Baseline results

The metrics for the two models from a baseline created by information retrieval techniques TF-IDF and cosine similarities, and tokenized by each model’s respective tokenizers are below:

Table 5.3: Comparison of the metrics from the extractive summarization tokenized by mBART and ClinicalT5

Metric	mBART	ClinicalT5
BLEU	0.012	0.011
ROUGE-1	0.270	0.269
ROUGE-2	0.142	0.141
ROUGE-L	0.158	0.156
BERTScore	0.125	0.127
METEOR	0.143	0.140
Align	0.262	0.105
MedCon	0.157	0.156
Overall	0.138	0.138

Both baselines show similar statistics for all the metrics. The overall low scores imply that we should implement a more complex preprocessing procedure to improve their performance. Perhaps, advanced information retrieval techniques and splitting input text into hospital course

and discharge instructions with more complex regular expression methods could help improve the baseline results.

5.1.4 Comparisons between models

This subsection compares the two models performance across all the metrics and against the baseline generated from information retrieval techniques. The results for both models indicate a lowest overall score compared to other (non)pre-processing methods, showing that the information retrieval approach is insufficient to generate quality summaries

Non-preprocessed text

mBART shows significant improvements in all metrics, notably the BLEU, ROUGE-1, and BERTScore, indicating better qualities for the generated summaries. ClinicalT5 also shows improvement, although not as large as mBART with BLEU increasing the most (0.0011 to 0.031)

Sentence Simplification

With the mBart model, BLEU, ROUGE-1, and BERTScore are the metrics that improve the most compared to the baseline. This indicates that this method allows coherent and relevant summary generation. Similarly, ClinicalT5 shows improvements in BLEU and ROUGE-1 scores, benefiting from this method.

Keyword Extraction

mBART shows even greater improvements on BLEU, ROUGE-1, and ROUGE-2, highlighting the importance of including a certain number of keywords in enhancing the qualities of the generations. However, ClinicalT5 shows a decline in the metrics, indicating a mismatch compared to other methods.

Entity Recognition

Both models show improvements in ROUGE-1 and BERTScore, showing that the generated summaries are contextually accurate

Overall, mBART is much more effective in medical text summarization compared to ClinicalT5, across all input preprocessing methods, as shown in the evaluation metrics - especially BLEU, ROUGE, and BERTScore. This means that these preprocessing methods are more important for improving the quality of the generated summaries. On the other hand, for ClinicalT5, not all preprocessing methods show improvements in model performance.

5.1.5 Other participants results

The final leaderboard for the challenge can be viewed at this [Google Sheet](#). The number one ranked team, "WisPerMed" has an overall score of 0.331, while the lowest ranked team - "AIMI-Baseline" has the lowest overall score of 0.102. The most significant difference can be observed from the BLEU score, with the first-ranked team being 10 times higher. The overall results from the mBART model, are between the 6th and 10th place, depending on the pre-processing methods (from 0.248 to 0.279). We can observe from Table 5.1 and the standings that while most of the metrics are similar to the equivalent teams in the standings, the ROUGE-1 scores are slightly lower and the AlignScore is slightly higher. The corresponding results for ClinicalT5 rank between 12th and 15th place, also varying on the pre-processing methods (from 0.150 to 0.188). The detailed metrics are generally similar to the other equally ranked teams, however, the METEOR is significantly lower (around 0.15 vs. 0.3) and the AlignScore is better (around 0.33 vs. 0.18).

5.2 AI and Human Evaluation

The semantic similarity between generated and reference summaries using Clinical-BERT and cosine similarity shows the quality of the generated summaries. This section provides statistics for both models with different text preprocessing methods.

5.2.1 Semantic Evaluation using clinicalBERT

ClinicalT5 results

- **Base Method (With no text preprocessing):** For the hospital course, the mean similarities across 1000 randomly selected records is 0.8448, with a standard deviation of 0.1135. This means that on average the generated texts align well with the inputs, however, some texts have inconsistencies. The generated discharge instructions share a slightly increased mean value (0.8636) with a lower standard deviation (0.0840). This implies better alignment and consistency to the reference text.
- **Sentence Simplification:** The average similarity of 0.8907 and standard deviation of 0.0860 shows that the texts are more aligned and consistent compared to the base method. For discharge instruction, a slight decrease in the mean (0.8280) and standard deviation (0.0420) means the generated texts are consistently worse than the base method.
- **Keyword Extraction:** For the hospital course, this method also shows an improvement, although not as big as the previous one, as shown in the average similarity of 0.8731 and standard deviation of 0.0860. On the other hand, the discharge instructions generated are on average worse than the base method (0.7497), however, this performance is subjected to low variability (0.0853)
- **Entity Recognition:** Compared to the base, the generated hospital course demonstrates an average similarity of 0.8448 but is more consistent across the sample (std of 0.1135). Conversely, for discharge instructions, the quality of the generated text is improved with an average similarity of 0.864 with the same consistency (std of 0.091)

mBART results

The semantic similarity for mBART is higher than ClinicalT5 across all preprocessing methods. Additionally, the former shows consistency across the generated summaries. The key points are discussed below.

- **Base Method (With no text preprocessing):** There are strong and consistent alignments between the generated and the reference text, as shown in The average similarity of 0.9641 and a low standard deviation of 0.0262 for hospital course and 0.948 and 0.037 for discharge instructions, highlighting mBART ability to maintain text similarity.
- **Sentence Simplification:** This method also demonstrates strong and consistent alignment with an average value of over 0.945 and a low std of around 0.03 for both hospital course and discharge instructions.
- **Keyword Extraction and Entity Recognition:** Both methods give mean results that are on par with the base and have the same consistencies as the Sentence Simplification method.

We can see that the mBART model’s generated texts remain more consistent after preprocessing methods, suggesting its superiority and adaptability in text summarization, whereas ClinicalT5 struggles both with similarities and consistency, especially with keyword extraction input. Pre-processing methods

The detailed statistics about the semantic evaluations for both models are in Table A.1 and A.2

5.2.2 Human Evaluation

As discussed in the previous section, the reference and the generated text will be judged on accuracy, clarity, completeness, and relevance. The results for the 3 randomly selected records are detailed below.

ClinicalT5 model

The sample evaluations for the hospital course are presented in Table 5.4.

Table 5.4: ClinicalT5 - Comparison of Generated and Reference Hospital Course with Scores

Generated Text	Reference Text
--- with pmhx CAD s/p multiple stents, chf EF 35-40% icd/ppm, HTN, vascular disease presents to the ED due to elevation of cr on outside labs.	Ms. --- is a --- year old female with ESRD --- DM and HTN (on HD, --- via right-sided AVF), DM2, CAD s/p CABG, HFpEF, HTN, and recurrent polymicrobial UTIs in the past (failed suppressive fosfomycin therapy in --- who presented with abdominal pain, nausea/vomiting, diarrhea and dysuria most likely due to viral gastroenteritis and UTI.
Score: Accuracy: 4, Clarity: 3, Completeness: 3, Relevance: 4. The generated text lacks detail and clarity compared to the reference, however, the relevant details are captured.	
--- year-old --- woman, past medical history of cerebral amyloid angiopathy with prior frontal ICH and s/p L ACA aneurysm clippin, who presents with nontraumatic left frontal intraparenchymal hemorrhage, a small subdural hematoma, bilateral intraventricular extension, 1 mm midline shift, and a prior bilateral anterior frontal lobe encephalomalacia, who presents with seizure and iph.	The patient was admitted to the colorectal surgery service and initiated on antibiotics. He was added on to the OR schedule for incision and drainage of perirectal abscess. He tolerated the procedure well without complication. After his incision and drainage, he was transferred to the floor, diet was advanced to regular diet and he was transitioned to PO antibiotics. Due to his profound leukocytosis (WBC of 18 on admission) he was discharged on a 7 day course of PO antibiotics.
Score: Accuracy: 4, Clarity 2, Completeness: 3, Relevance: 3. The generated text captures the patient's conditions but fails to provide procedures and outcomes details.	

Sample evaluations for discharge instructions are in table 5.5, the examples are shortened for clarity and ease of reading.

mBART model

The sample evaluations for hospital course is shown in table 5.6.

Compared to ClinicalT5 model, mBart generates less details and is not as coherence. Moreover, the model lacks comprehensive information and tends to be more repetitive.

The sample evaluations for discharge instructions is presented in table 5.7.

Compared to ClinicalT5, the model's overall text quality is considerably better in all four criteria. There is no repetition of discharge instructions in mBART because of no token padding, meaning that the model manages to capture critical information. The generated text also contains special characters such as which ClinicalT5 failed to capture. This section will address the research questions, the advantages and limitations, and the potential applications of this study.

Table 5.5: ClinicalT5 - Comparison of Generated and Reference discharge instructions with Scores and Comments

Generated Text	Reference Text
Dear Mr. ___, It was a pleasure taking care of you at the ___ ___. You were admitted to the ___ ___ for further evaluation. You were admitted to the ___ ___ for further evaluation. You were admitted to the ___ ___ for further evaluation.	Dear Ms. ___, You were admitted to ___ because you were having abdominal pain, nausea, vomiting, and diarrhea. Your symptoms were most likely caused by a urinary tract infection. You may also have had a viral stomach bug causing your diarrhea. You were treated with antibiotics for your urinary tract infection. You will need to continue antibiotics after discharge to complete the treatment course. You also became dizzy while you were here. This is most likely benign paroxysmal positional vertigo (BPPV) which is not dangerous. Be sure to change positions slowly and drink plenty of fluids at home. You missed a dialysis session at home before admission...
Score: Accuracy: 1, Clarity: 1, Completeness: 1, Relevance: 1. Comments: The generated text is repetitive and fails to represent the details of the reference text.	

Table 5.6: mBART - Comparison of Generated and Reference Hospital Course with Scores and Comments

Generated Text	Reference Text
___ with PMHx CAD s/p multiple stents, CHF EF 35-40% ICD/PPM, HTN, vascular disease presents to the ED due to elevation of Cr on outside labs. Elevated Cr: Cr on ___ was 7.2 so he was sent in for further evaluation by his nephrologist Dr. _____. Patient reports that he had seen his PCP 2 weeks ago and he had a weight of approx. 180 lbs (up from his dry weight). His PCP changed his medication to torsemide and he has dropped to 162 lbs now. Per conversation with patient's PCP, Cr has been up-trending over the past several weeks. On ___, he presented to PCP with significant lower extremity edema and was started on Torsemide 60 mg and spironolactone 25 mg. Home potassium chloride and furosemide were stopped at this time. He was referred into the ED. Given that Cr was elevated to 7.2 on check today, the patient was referred to the ED. He denied fevers, chills, chest pain, SOB, or any other symptoms. He denies increased lower extremity edema. His UA color straw appear clear specGR 1.010 pH 6.	Ms. ___ is a ___ year old female with ESRD ___ DM and HTN (on HD, ___ via right sided AVF), DM2, CAD s/p CABG, HFpEF, HTN, and recurrent polymicrobial UTIs in the past (failed suppressive fosfomycin therapy in ___) who presented with abdominal pain, nausea/vomiting, diarrhea and dysuria most likely due to viral gastroenteritis and UTI. Acute complicated cystitis Presenting symptoms including suprapubic abdominal discomfort, dysuria and urinary frequency with positive urinalysis on arrival were consistent with recurrent UTI. Infection considered complicated given diabetes. There was no CVA tenderness or findings on CT scan to raise concern for pyelonephritis. Urine culture eventually grew pansensitive E. Coli as well as enterococcus which was sensitive to ampicillin. She was initially empirically treated with Unasyn then transitioned to Augmentin after urine culture sensitivities returned. Abdominal pain and dysuria improved with treatment. She will need to complete a 10 day course of Augmentin after discharge (___)...
Score: Accuracy: 3, Clarity: 3, Completeness: 3, Relevance: 4. Comments: The generated text has several key details such as medical conditions but is missing detailed observations.	
Ms. ___ is a ___ year-old R-handed woman, past medical history of cerebral amyloid angiopathy with prior frontal IPH and s/p L ACA aneurysm clipping, who presents with seizure and IH. IH: Patient presented with a left frontal intraparenchymal hemorrhage. She was intubated for airway protection with Etomidate 20, succinylcholine 125, and propofol. Her noncon head CT revealed a small subdural hematoma, bilateral intraventricular extension, 1 mm midline shift, and a prior bilateral anterior frontal lobe encephalomalacia. There is no family at bedside, so history is per chart review. The patient was evaluated by neurosurgery who stated there is no urgent neurosurgical intervention. She was given 2 mg of ativan...	The patient was admitted to the colorectal surgery service and initiated on antibiotics. He was added on to the OR schedule for incision and drainage of perirectal abscess. He tolerated the procedure well without complication. After his incision and drainage, he was transferred to the floor, diet was advanced to regular diet and he was transitioned to PO antibiotics. Due to his profound leukocytosis (WBC of 18 on admission) he was discharged on a 7 day course of PO antibiotics. At the time of discharge, pain was controlled on oral pain medication, he was out of bed to ambulate without assistance,...
Score: Accuracy: 4, Clarity: 4, Completeness: 4, Relevance: 4. Comments: This is a detailed generated text, matching the reference on patient's condition and treatment.	

Table 5.7: mBART - Comparison of Generated and Reference Discharge Instructions with Scores and Comments

Generated Text	Reference Text
Dear Mr. ---, It was a pleasure taking care of you at --- ---. You were admitted to the hospital because you had increased fluid in your legs and lower extremity. You were found to have a urinary tract infection. We treated you with antibiotics and your symptoms improved. Please follow up with your primary care doctor and nephrologist as listed below. We wish you the best! Sincerely, Your --- Care Team	Dear Ms. ---, You were admitted to --- because you were having abdominal pain, nausea, vomiting and diarrhea. Your symptoms were most likely caused by a urinary tract infection. You may also have had a viral stomach bug causing your diarrhea. You were treated with antibiotics for your urinary tract infection. You will need to continue antibiotics after discharge to complete the treatment course... Please take your medications as prescribed and attend all follow-up appointments as scheduled. It was a pleasure participating in your care - we wish you all the best. Sincerely, Your --- Team
Score: Accuracy: 3, Clarity: 4, Completeness: 3, Relevance: 4. Comments: The generated text lacks follow-up and additional health advice. However, it manages to capture the patient's summary and conditions.	
Dear Ms. ---, It was a pleasure taking care of you at --- ---. You were admitted to the hospital because you were having seizures and were found to have a small bleed in your left frontal hemorrhage. You had a CT scan of your brain which did not show any bleeding. You also had an MRI of your head which was normal. You were evaluated by the neurosurgery team who did not feel that there was any need for surgery. You will need to follow up with your primary care physician as an outpatient. We wish you the best, Your --- Neurology Team	You have undergone a surgical procedure in the office for the drainage of a perirectal abscess. We suggest that you go home, get off your feet and do as little as possible in the way of physical activity for the rest of the day. An ice bag against the rectal area may be helpful. The numbing medicine that we use generally wears off in 1 ½ to 2 ½ hours and at that time, you will experience some pain. You may have been given a prescription for pain, use it if you need to.... You may be experiencing a complication of your surgery. • General increased rectal pain • Fever of 101°F or higher • Difficulty urinating • Large amount of bleeding that does not stop
Score: Accuracy: 4, Clarity: 4, Completeness: 3, Relevance: 4. Comments: The generated covers the main points but is missing several details such as specific follow-up or potential complications.	

Chapter 6

Discussion

The experiments provide insights into how a variant of the transformer-based model - ClinicalT5 and a combination of BERT and GPT models - mBART performs on medical and radiology text summarization. This focuses on how different NLP preprocessing methods can affect the text quality.

6.1 Discharge Me! Dataset

The dataset is based on the MIMIC-IV, which is a large anonymized dataset of admitted patients to the intensive care unit or emergency department at the Beth Israel Deaconess Medical Center in Boston, MA [13]. It is featured in the study with a primary objective of generating comprehensive hospital courses and discharge instructions. Throughout the experiments, it shows important attributes to evaluate the performance of the mBART and ClinicalT5 models.

6.1.1 Complexity and Diversity of Clinical Text

The dataset offers over 100000 records, which likely covers a broad range of medical specialties, such as cardiology and neurology. Inside these, they include a broad range of terminology and contexts. Another valuable attribute of this dataset is that it composes a variety of demographics, including patient's ages, genders, and health conditions. These attributes mean that the trained models, if capable, can adapt to different clinical scenarios, enabling them for real-world application.

Within the dataset, the clinical records consist of complex medical acronyms, jargon, and abbreviations; along with their unstructured nature with various formatting styles. All of them are challenging for both models to learn the patterns in the data. Overall, these properties require complex pre-processing methods and capable models to ensure accurate summaries.

6.1.2 Clinical records' length and level of detail

The clinical notes vary greatly in length, as can be seen in Figure 4.1, from short records to detailed descriptions. As a result, they can be in different levels of detail, from high-level assessments to detailed medical histories. This means that besides the complex and diverse nature of the clinical text, the models also need to consider balancing between concise and complete summaries by making sure important information is kept.

6.1.3 Impacts of token constraints and preprocessing methods

Both models' token limits are set to 512, which is the maximum ClinicalT5 can handle. From Figures 6.1 and 6.2, it can be seen that most of the tokenized hospital courses and discharge instructions contain 512 tokens. This means that they usually exceed the tokens limit, leading

```
[ ] summary_lengths = [len(tokenizer.encode(summary, max_length=512, truncation=True)) for summary in summaries_train_course]
print(f"Max summary length: {max(summary_lengths)}")
# Based on this information, update max_target_length accordingly
optimal_max_target_length = int(np.percentile(summary_lengths, 95))
print(f"95th percentile summary length: {np.percentile(summary_lengths, 95)}")
```

Figure 6.1: Hospital Course 95th percentile length in tokens

```
summary_lengths = [len(tokenizer.encode(summary, max_length=512, truncation=True)) for summary in summaries_train_instructions]
print(f"Max summary length: {max(summary_lengths)}")
# Based on this information, update max_target_length accordingly
optimal_max_target_length = int(np.percentile(summary_lengths, 95))
print(f"95th percentile summary length: {np.percentile(summary_lengths, 95)}")
```

Max summary length: 512
95th percentile summary length: 512.0

Figure 6.2: Discharge Instructions 95th percentile length in tokens

to potential loss of information. Although methods such as keyword extraction and entity recognition could alleviate this limitation by moving important information at the front, they vary in effectiveness; as can be seen in the ClinicalT5 under the keyword extraction method, experiencing a significant drop in overall metrics, or a drastic increase with the same method for mBART.

6.2 ClinicalT5 vs. mBART performance

6.2.1 ClinicalT5

- **With NLP methods:** The model’s performance when including NLP methods such as entity extraction and sentence simplification is improved. For example, the model trained on text with medical entities achieved a lower train and validation loss compared to the base. The texts retain the most important information, in exchange for the slight reduction in literature quality.
- **Without NLP methods:** The performance of the model is less desirable, as indicated in higher train and validation losses. The generated texts align less with the reference and sometimes miss important details.

6.2.2 mBART

BART performs considerably better than ClinicalT5 across all NLP methods, as shown in the evaluation metrics, as well as human and AI evaluation.

- **With NLP methods:** The model also experienced performance gain, specifically with keyword extraction and entity recognition, which both contributed to a better text quality
- **Without NLP methods:** The drop in performance is more noticeable in mBART, this implies that mBART relies more on input preprocessing to maximize its capabilities against the intricacies of medical texts.

6.2.3 Best input representation

The optimal preprocessing methods depend on the model architecture and training process, which the experiment performed inadequately due to a lack of computing resources. mBART benefits more from keyword extraction, indicating that it can benefit from well-structured and focused input. ClinicalT5, on the contrary, showed mixed results for different preprocessing methods.

The best input representation is possibly mBART with keyword extraction, emerging as the most reliable summarization method in this experiment.

6.2.4 Other team approach

The first ranked team "WisPerMed" implements various methods for their submissions, including few-shot learning, instruction tuning, MIMIC section identification, Dynamic Expert Selection (DES), and an additional clinical dataset utilization; where the combination of DES with deep learning methods give the best performance. Specifically, the few-shot learning method uses the WizardLM-2 model on a few prompts to generate the desired sections. The instruction tuning method uses "diverse instruction-based datasets to train the model", also with different prompts for the two target sections, with the involvement of the Asclepius dataset containing synthetical clinical notes and Instructions-Answer pairs. The DES method selects a set of models focused on different readability and factuality metrics to simulate expert evaluations [4].

The experiments, given additional time and research efforts, could adapt several strategies from the above work. First, the Asclepius dataset can be used to instruct tune both mBART and ClinicalT5 by adapting from a prompt from Figures 6 and 7 of the paper [4]. Then, the workflow from Figure 1 (MIMIC section identification) - calculating and sorting training text based on BERTScore can be applied to both target sections. Finally, we can construct different DES based on readability and factuality metrics to generate the submission data frame.

6.3 Handling complex language in Radiology Reports

Both models showed the potential to deal with the intricacies of radiology reports languages. However, in this study setting, the weaknesses outweigh the advantages:

- **Strengths:** These models understand and can generalize the radiology report. This is reflected by several instances of hospital courses that the key points are present in the summary, as shown in this excerpt: "Her noncon head CT revealed a small subdural hematoma, bilateral intraventricular extension, 1 mm midline shift, and a prior bilateral anterior frontal lobe encephalomalacia."
- **Weaknesses:** Due to the limitation of 512 tokens as inputs, the models struggle to capture the nuances of radiology reports, despite methods that put the important entities or keywords to the beginning. As a result, important details are sometimes overlooked thus the summaries lack depth for complete understanding.

6.4 Advantages, Limitations, and Applicability of the Studies

Advantages

Through the experiments, we see that the text summarized by both models can provide references, helping professionals with this daunting task. Moreover, most of the outputs from both models are consistent, addressing the variability that might stem from human-summarized texts (due to the large volume of text, fatigue, ...). Lastly, the model is capable of handling massive datasets, which is required for the creation of a Clinical Decision Support System (CDSS).

Limitations

The model outcome depends on imperfect text preprocessing methods, which might not always be flexible due to intricacies and nuances in medical language. Fine-tuning the models thus requires additional domain-specific data, which is not readily available. Second, the training and evaluation of these models are computationally heavy and expensive; before access to the Radboud cloud cluster was granted, the models were trained on Google Colab's GPU, costing €11.99 for 100 units. This might be a drawback for several clinics that are not adequately funded. Third, the study could have benefited by more board datasets, covering larger medical domains. Lastly, opinions from professionals through real-world clinical settings could offer more well-rounded evaluations of the generated text.

Real-life applications

The trained ClinicalT5 and mBART models can be integrated into clinical systems such as electronic health records (EHR) to automate patient full visits by providing discharge summaries and hospital courses, facilitating communications between doctors and patients on complex medical matters. Moreover, researchers can baseline these models to go over a vast amount of medical and radiology literature, assisting in pinpointing key information and trends.

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Appendix A

Appendix

A.1 Training Loss, Accuracy and Validation Loss results for mBART and ClinicalT5

A.1.1 mBART

Figure A.1 shows that, the models experience a steep drop in training loss - then stabilized, indicating faster learning and no overfitting. The entities method showed the most promising performance for both hospital course and discharge instructions generation, followed by sentence simplification. On the other hand, keywords extraction reveals the highest losses.

In figure A.4, during the first epoch, we can see greater fluctuations in the validation loss. After that, most models show less fluctuations, indicating more consistent performances. Similar to training loss, sentence simplification and entities recognition exhibits a consistently low values and keywords extraction shows the highest loss.

A.1.2 ClinicalT5

Figure A.3, similar to mBart, shows that all models experience a drastic drop in train loss from the first epoch, indicating that model quickly learns the data pattern. Then, the loss stabilized and the curve flattened, indicating no overfitting. We can see that the model trained on text with medical entities and sentence simplification experience the lowest loss comparing to the others pre-processing methods (keywords extraction and no preprocessing). We should consider fine-tuning this method to improve the performance further.

From figure A.6, unlike the training loss, model trained on input with sentence simplification exhibits the lowest validation loss value during the epochs. Also, the instructions data give significantly lower loss than course data, showing easier contents in the input.

Table A.1: Results for the mBART model on inputs of 1024 tokens with various preprocessing methods. Here, R stands for the ROUGE metric, BERT is for BERTScore and MET is for METEOR

Preprocessing Method	BLEU	R-1	R-2	R-L	BERT	MET	Align	Med	Overall
Base (no preprocessing)	0.032	0.316	0.130	0.194	0.270	0.184	0.173	0.213	0.189
Sentence Simplification	0.078	0.379	0.144	0.240	0.371	0.256	0.286	0.329	0.260
Keywords Extraction	0.076	0.385	0.192	0.246	0.363	0.249	0.285	0.316	0.264
Entity Recognition	0.078	0.368	0.163	0.219	0.371	0.253	0.280	0.325	0.257



Figure A.1: Training loss for mBART model



Figure A.2: Training loss for mBART model with inputs of 1024 tokens

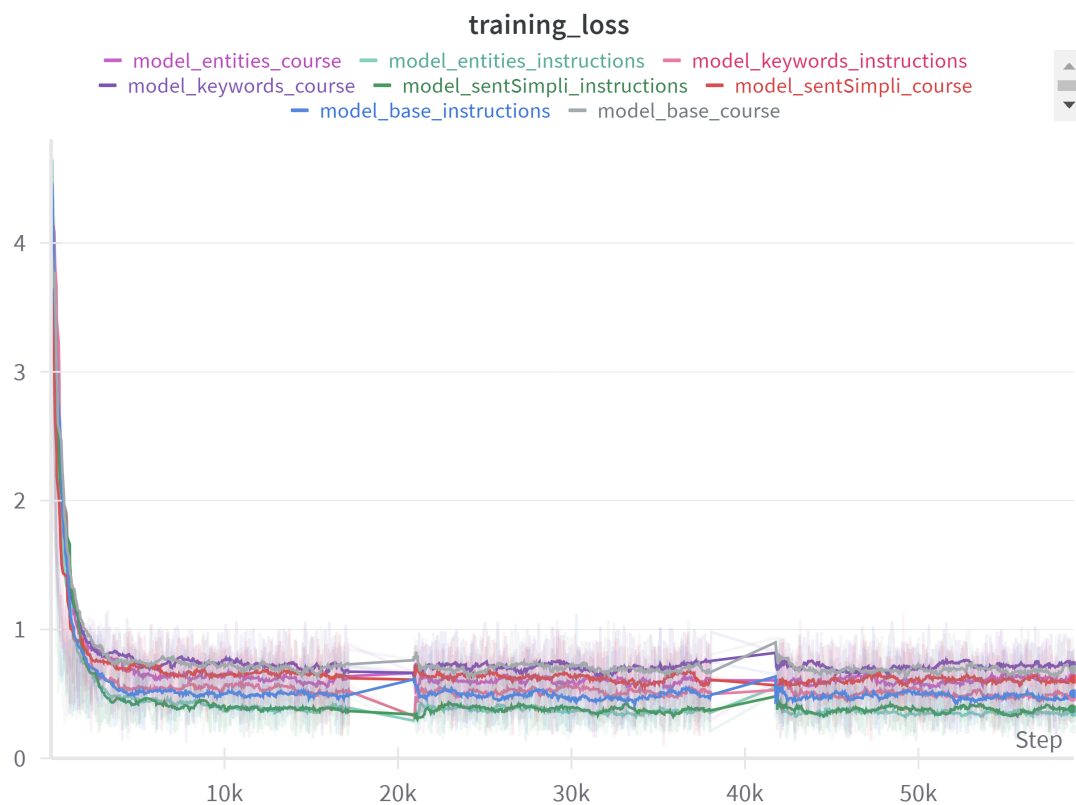


Figure A.3: Training loss for clinicalT5

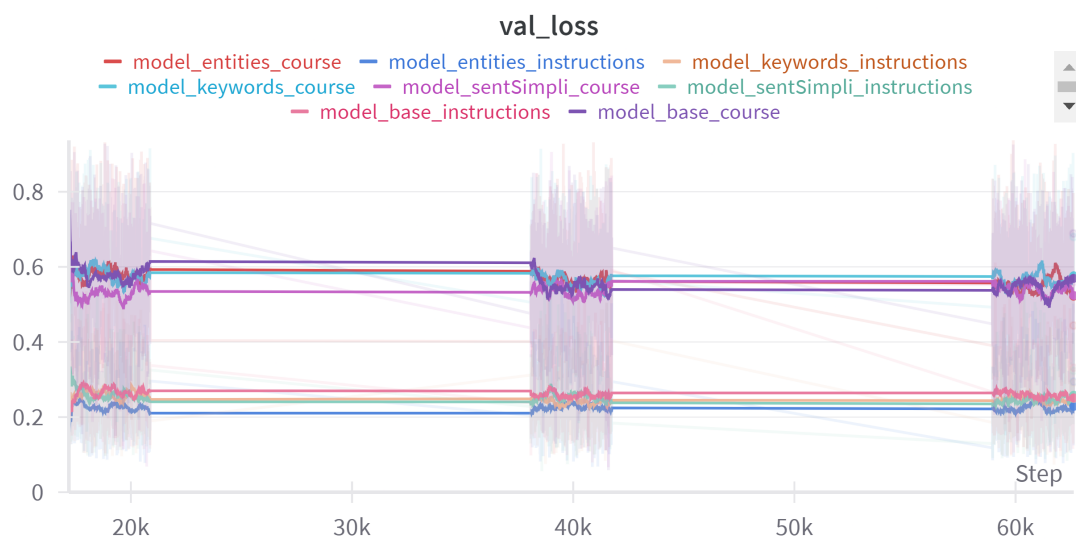


Figure A.4: Validation loss for mBART

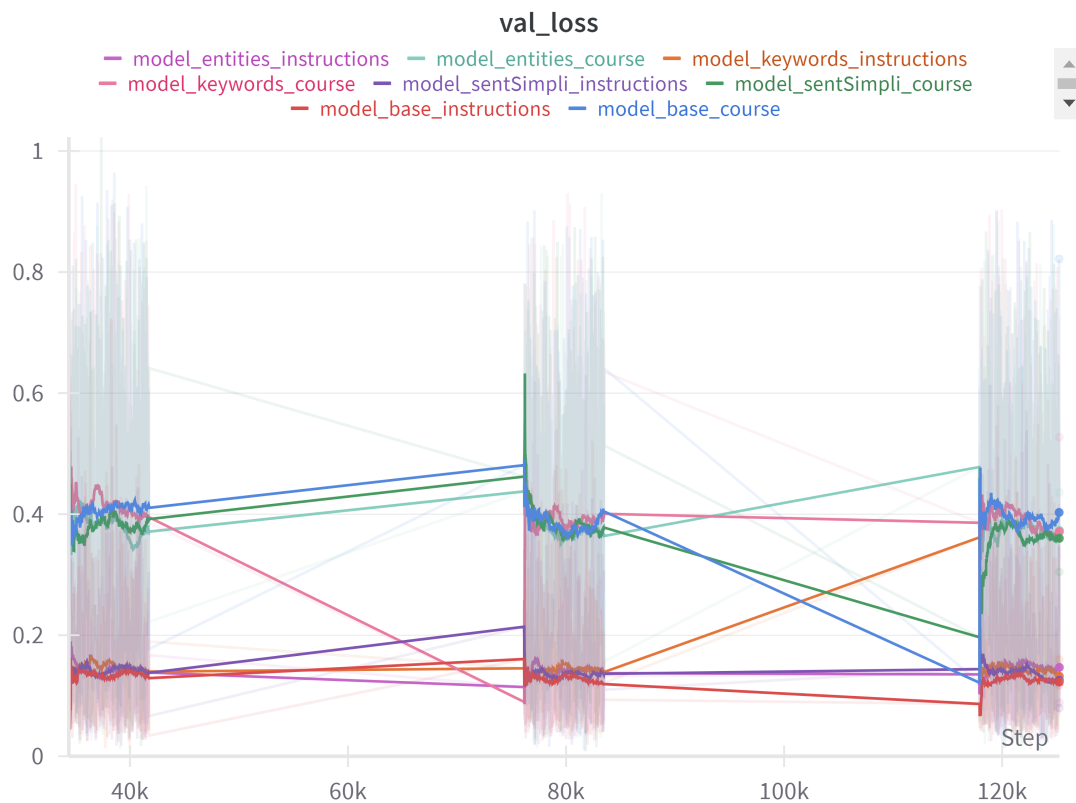


Figure A.5: Validation loss for mBART with inputs of 1024 tokens

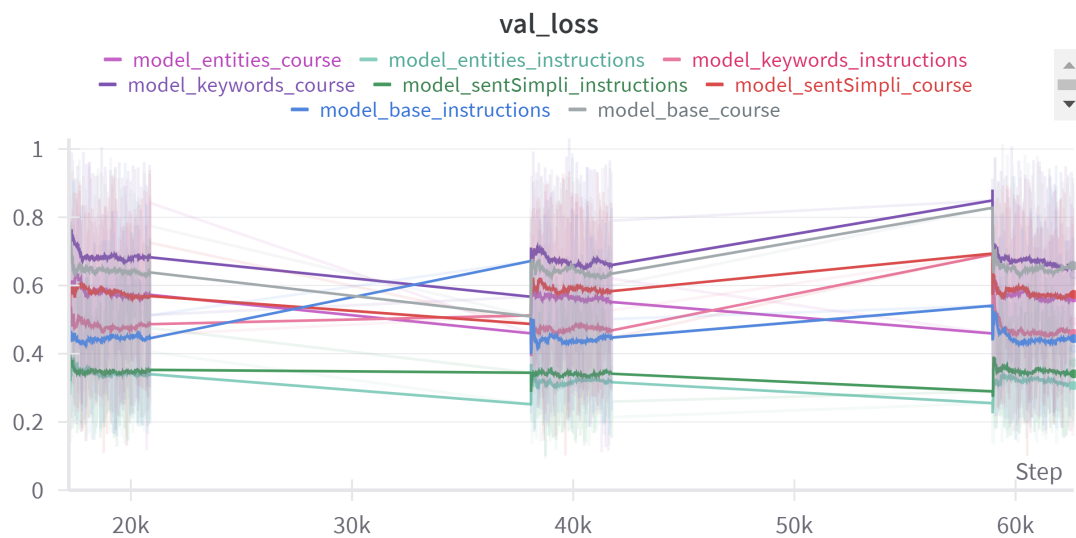


Figure A.6: Validation loss for clinicalT5

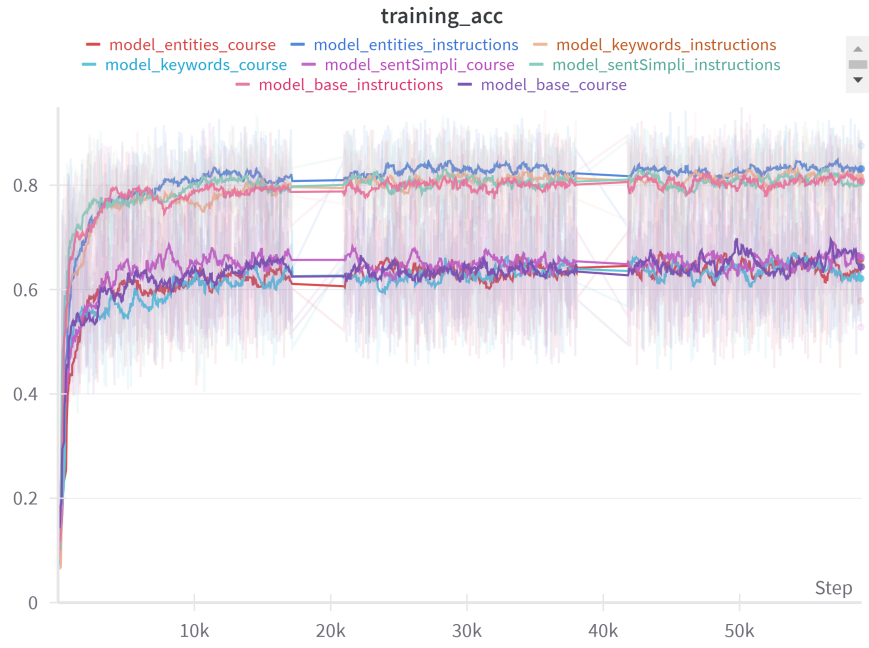


Figure A.7: Training accuracy for mBART model



Figure A.8: Training accuracy for mBART model with inputs of 1024 tokens



Figure A.9: Training accuracy for ClinicalT5 model

A.2 Statistics for Semantic Evaluation using ClinicalBERT for both models

Table A.2: Statistics for Semantic Evaluation using ClinicalBERT for Different Pre-processing Methods on ClinicalT5 Model

Pre-processing Method	Hospital Course Similarity				Discharge Instructions Similarity			
	Mean	Std	Min	Max	Mean	Std	Min	Max
Base Method	0.8448	0.1135	0.3653	0.9979	0.8636	0.0840	0.3497	0.9795
Sentence Simplification	0.8907	0.0860	0.3027	0.9988	0.8280	0.0420	0.5682	0.9384
Keyword Extraction	0.8731	0.0661	0.5212	0.9641	0.7497	0.0853	0.4049	0.9238
Entity Recognition	0.8448	0.1135	0.3653	0.9979	0.8636	0.0840	0.3497	0.9795

Table A.3: Statistics for Semantic Evaluation using ClinicalBERT for Different Pre-processing Methods on mBART Model

Pre-processing Method	Hospital Course Similarity				Discharge Instructions Similarity			
	Mean	Std	Min	Max	Mean	Std	Min	Max
Base Method	0.9461	0.0262	0.6353	0.9942	0.9476	0.0371	0.5524	0.9959
Sentence Simplification	0.9474	0.0248	0.6862	0.9937	0.9494	0.0331	0.5635	0.9936
Keyword Extraction	0.9461	0.0261	0.6307	0.9911	0.9467	0.0362	0.5667	0.9928
Entity Recognition	0.9465	0.0252	0.6369	0.9941	0.9507	0.0294	0.5549	0.9951

A.3 Code repository

The full repository for the project can be found at: https://github.com/lamngue/Masters_Thesis